Black–Scholes & the Greeks

The BSM model turns five observable inputs into a fair option price and a full risk report — shown through QQQ (Nasdaq 100) options.

The Black–Scholes–Merton (BSM) model turns five observable inputs — spot, strike, time, rate, volatility — into a fair option price *and* a full risk report (the Greeks). Fifty years on, it remains the language in which options are quoted, hedged, and risk-managed.

PROJECT CARD

CATEGORY

Quant Foundations & Derivatives

LIBRARIES

NumPy · SciPy · Matplotlib

ASSETS

QQQ options (Nasdaq 100 ETF, Invesco)

TIMEFRAME

Jan 2018 – Dec 2024 (trailing 1y vol)

USE

Pricing & hedging options; implied vol is how the entire options market quotes prices

1. Summary

BSM assumes the underlying follows **Geometric Brownian Motion** (see the previous tutorial) and shows that an option's payoff can be **replicated** by continuously trading the stock and a bond — so the option's price is the cost of that replication, independent of anyone's market view. The result is a closed-form price for European calls and puts, and analytic **Greeks**: the sensitivities of that price to spot (Δ, Γ), volatility (ν, "vega"), time (Θ) and rates (ρ).

2. Intuition

2.1 Replication, not prediction

Hold Δ shares against a short call and the portfolio is (momentarily) immune to small price moves. Rebalance continuously and the option is manufactured from stock and cash. No-arbitrage then pins the price — the expected return of the stock μ **drops out entirely**. That is the magic: two people who disagree wildly about where the Nasdaq is going must still agree on the option's fair price.

2.2 Volatility is the price of uncertainty

Of the five inputs, four are observable. Volatility is not — and the option premium is, in essence, the market's bid on future uncertainty. Higher σ → wider terminal distribution → both calls and puts gain value (optionality only benefits from dispersion).

2.3 Greeks: the risk dashboard

A trader rarely asks "what is the price?" — they ask "what happens to my book if spot drops 1%, vol jumps 2 points, and a day passes?" Delta, Gamma, Vega and Theta answer exactly that, one partial derivative at a time.

Greek	Sensitivity to	Typical use
Delta Δ	Spot	Hedge ratio
Gamma Γ	Spot (2nd order)	Hedge stability / convexity
Vega ν	Volatility	Vol exposure
Theta Θ	Time	Daily carry / decay
Rho ρ	Interest rate	Rate exposure

3. Theory & Mechanics

3.1 The pricing formula

With spot S, strike K, maturity T, rate r and volatility σ:

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

$$C = S\,N(d_1) - Ke^{-rT}N(d_2), \qquad P = Ke^{-rT}N(-d_2) - S\,N(-d_1)$$

Read $N(d_2)$ as the (risk-neutral) probability the option finishes in the money; $N(d_1)$ is the delta of the call.

3.2 The Greeks in closed form

$$\Delta_C = N(d_1), \quad \Gamma = \frac{N'(d_1)}{S\sigma\sqrt{T}}, \quad \nu = S\,N'(d_1)\sqrt{T}, \quad \Theta_C = -\frac{SN'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$$

```python
def bs_price(S, K, T, r, sigma, kind="call"):
    """Black-Scholes price for a European call or put."""
    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    if kind == "call":
        return S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
    return K * np.exp(-r * T) * norm.cdf(-d2) - S * norm.cdf(-d1)

def bs_greeks(S, K, T, r, sigma, kind="call"):
    """Delta, Gamma, Vega, Theta (per year), Rho for a European option."""
    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    sign = 1 if kind == "call" else -1
    return {
        "delta": sign * norm.cdf(sign * d1),
        "gamma": norm.pdf(d1) / (S * sigma * np.sqrt(T)),
        "vega":  S * norm.pdf(d1) * np.sqrt(T),
        "theta": (-S * norm.pdf(d1) * sigma / (2 * np.sqrt(T))
                  - sign * r * K * np.exp(-r * T) * norm.cdf(sign * d2)),
        "rho":   sign * K * T * np.exp(-r * T) * norm.cdf(sign * d2),
    }
```

3.3 Put-call parity — the free sanity check

Independent of any model: $C - P = S - Ke^{-rT}$. If your implementation violates it, something is wrong.

4. Applied Example — QQQ

4.1 Inputs from real data

Spot from the latest QQQ close; volatility proxied by the trailing 1-year realised vol of log returns (in practice you would use the implied vol quoted in the market — that distinction is exactly what the *Heston vs Black-Scholes* piece explores).

```python
TICKER = "QQQ"
START, END = "2018-01-01", "2024-12-31"

def load_prices(ticker, start, end):
    """Adjusted-close prices: yfinance first, Stooq as fallback."""
    try:
        import yfinance as yf
        df = yf.download(ticker, start=start, end=end, auto_adjust=True, progress=False)
        if not df.empty:
            return df["Close"].squeeze().rename(ticker).dropna()
    except Exception as exc:
        print(f"yfinance failed ({exc}); trying Stooq…")
    url = f"https://stooq.com/q/d/l/?s={ticker.lower()}.us&i=d"
    df = pd.read_csv(url, parse_dates=["Date"], index_col="Date")
    return df.loc[start:end, "Close"].rename(ticker).dropna()

px = load_prices(TICKER, START, END)
log_ret = np.log(px / px.shift(1)).dropna()

S0    = float(px.iloc[-1])
SIGMA = float(log_ret.tail(252).std() * np.sqrt(252))   # trailing 1y realised vol
R     = 0.045                                           # short-term risk-free rate
T3M   = 0.25                                            # 3 months
K_ATM = round(S0)                                       # at-the-money strike

call = bs_price(S0, K_ATM, T3M, R, SIGMA, "call")
put  = bs_price(S0, K_ATM, T3M, R, SIGMA, "put")
```

```python
print(f"{TICKER} spot = {S0:,.2f}   sigma = {SIGMA:.2%}   r = {R:.2%}")
print(f"3M ATM call (K={K_ATM}): {call:6.2f}")
print(f"3M ATM put  (K={K_ATM}): {put:6.2f}")

g = bs_greeks(S0, K_ATM, T3M, R, SIGMA, "call")
print("\n3M ATM call Greeks:")
print(f"  delta {g['delta']:7.3f}   gamma {g['gamma']:8.5f}   vega {g['vega']:7.2f}"
      f"   theta {g['theta']/365:7.3f}/day   rho {g['rho']:6.2f}")
```

4.2 The Greeks across strikes

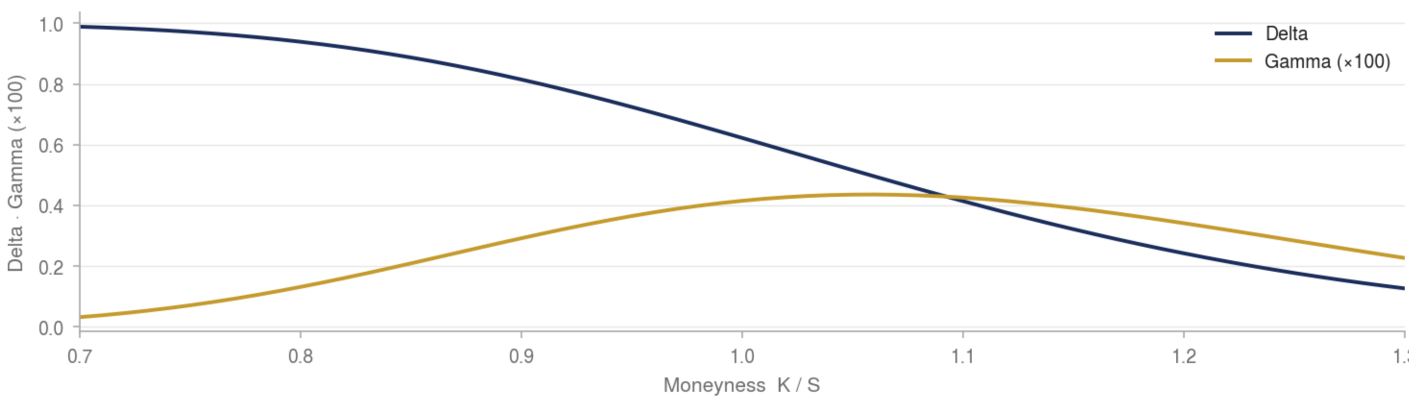
The same option book looks completely different in-, at- and out-of-the-money. Gamma and Vega **peak at the money** — that is where hedges churn fastest and vol exposure is largest; Delta rolls from 0 to 1 like a smoothed step function.

```python
strikes = np.linspace(0.80 * S0, 1.20 * S0, 200)
G = {k: [] for k in ["delta", "gamma", "vega", "theta"]}
for K in strikes:
    gk = bs_greeks(S0, K, T3M, R, SIGMA, "call")
    for k in G: G[k].append(gk[k])

fig, axes = plt.subplots(2, 2, figsize=(11, 6), sharex=True)
for ax, k in zip(axes.flat, G):
    ax.plot(strikes / S0, G[k], color="steelblue")
    ax.axvline(1.0, color="gray", ls=":", lw=1)
    ax.set_title(k.capitalize()); ax.set_xlabel("Moneyness  K / S")
fig.suptitle(f"{TICKER} 3M call Greeks across strikes  (S={S0:,.0f}, sigma={SIGMA:.0%})")
plt.tight_layout(); plt.show()
```

Figure 4.2 · QQQ 3-month call — the Greeks across strikes

Delta falls smoothly from 1 to 0 while Gamma, scaled by 100 to share the axis, peaks near the money



Source: Author calculations, from the article's computed results. Three-month QQQ call, S = 507.45, σ = 18.04%, r = 4%.

4.3 Time decay — the option's ticking clock

Repricing the ATM call as maturity shrinks shows Theta at work: decay is slow far from expiry and **accelerates sharply in the final weeks** — the reason short-dated option selling and 0DTE trading are games of Theta and Gamma.

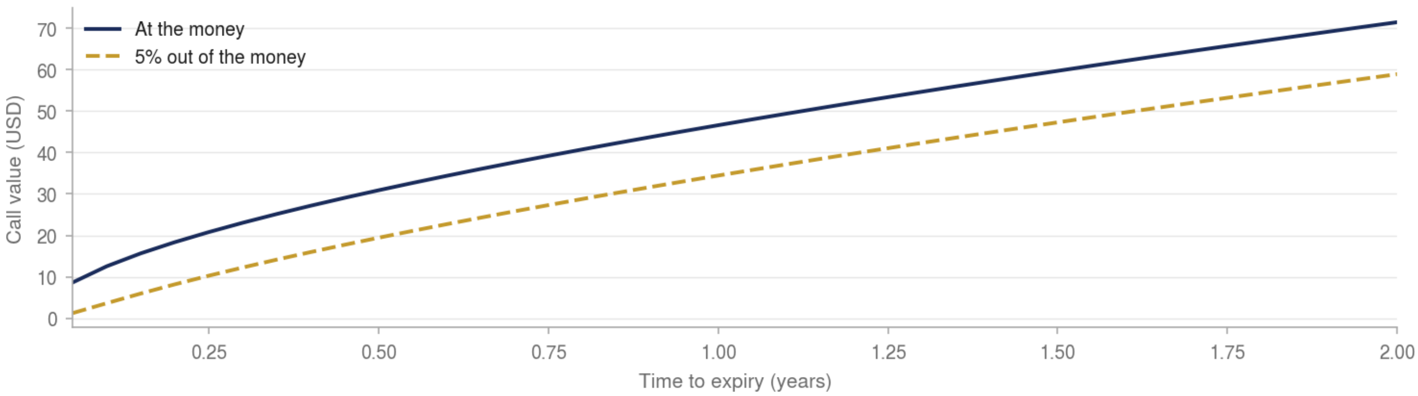
```python
maturities = np.linspace(1e-3, 1.0, 200)
atm_prices  = [bs_price(S0, K_ATM, t, R, SIGMA, "call") for t in maturities]
otm_prices  = [bs_price(S0, 1.05 * S0, t, R, SIGMA, "call") for t in maturities]

fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(maturities * 12, atm_prices, label=f"ATM (K={K_ATM})", color="steelblue")
ax.plot(maturities * 12, otm_prices, label="5% OTM", color="darkorange")
ax.set_xlabel("Months to expiry"); ax.set_ylabel("Call value")
```

```
ax.set_title(f"{TICKER}: value vs time to expiry — Theta accelerates near zero")
ax.invert_xaxis(); ax.legend()
plt.tight_layout(); plt.show()
```

Figure 4.3 · QQQ call value vs time to expiry — Theta accelerates near zero

At-the-money and 5% out-of-the-money calls; both decay toward zero and steepen in the final weeks



Source: Author calculations, from the article's computed results. Same parameters as above.

4.4 Sanity checks: parity and Monte Carlo

Two independent tests of the implementation:

- **Put-call parity** must hold to machine precision.
- **Monte Carlo under GBM** (the simulator from the previous tutorial, with drift r) must converge to the closed-form price — Black-Scholes *is* the GBM expectation in disguise.

```
# 1) put-call parity
lhs, rhs = call - put, S0 - K_ATM * np.exp(-R * T3M)
print(f"parity: C - P = {lhs:.6f}   S - K e^-rT = {rhs:.6f}   diff = {abs(lhs-rhs):.2e}")

# 2) Monte Carlo pricing under risk-neutral GBM
N = 200_000
z = np.random.normal(size=N)
S_T = S0 * np.exp((R - 0.5 * SIGMA**2) * T3M + SIGMA * np.sqrt(T3M) * z)
mc_call = np.exp(-R * T3M) * np.maximum(S_T - K_ATM, 0).mean()
se      = np.exp(-R * T3M) * np.maximum(S_T - K_ATM, 0).std() / np.sqrt(N)
print(f"Monte Carlo call: {mc_call:.3f} ± {2*se:.3f}   closed form: {call:.3f}")
```

5. Conclusion

5a. Strengths

- **Closed form** — instant prices and Greeks, no simulation needed
- **Preference-free** — the stock's expected return drops out; only volatility matters
- **The market's lingua franca** — options are *quoted* in BSM implied vol even where the model's assumptions fail
- **Analytic Greeks** — the entire hedging industry runs on them

5b. Weaknesses & Limitations

- **Constant volatility** — markets trade a *smile/skew*, not a flat vol (one number cannot price all strikes)
- **GBM underlying** — no jumps, no fat tails; deep OTM puts are systematically underpriced
- **European exercise only** — American-style early exercise needs trees or numerical methods
- **Continuous frictionless hedging** — real rebalancing is discrete and costs money

5c. Applications in Practice

- Pricing and hedging European options and warrants
- Implied volatility extraction — the quoting convention for the entire options market
- Real-time Greek dashboards for market-making and portfolio risk

5d. Alternatives & Extensions

- **Stochastic volatility** — Heston (see our Quant Insights piece *"Heston vs Black-Scholes: fitting the volatility smile"*)
- **Jump-diffusion** — Merton, for crash risk and short-dated skew
- **Binomial trees** — American exercise, dividends, path-dependence
- **Local volatility** — Dupire, fitting the entire observed surface